Lecture 14

Hydrogen: the Hamiltonian

Objectives. *We prove that the Hamiltonian of hydrogen is self-adjoint on the domain of the free energy. Hydrogen is stable: no state in this domain has an energy mean below* $-\mathrm{Ry}$*, and* $-\mathrm{Ry}$ *is an eigenvalue. The spectrum contains* $[0,\infty)$*. Below zero it is discrete, and every eigenvalue is negative. Lecture 15 finds these eigenvalues.*

- Control the Coulomb potential on $D(H_0)$.
- Prove the sharp lower bound and exhibit its ground eigenvector.
- Explain how the domain excludes the singular solution at the origin.

14.1 The Coulomb Hamiltonian

In a hydrogen atom an electron of mass m and charge $-e$ moves around a proton of charge e. We keep the proton fixed at the origin, and § 15.3 says what the motion of the proton changes. The potential energy of the electron at x is

$$V(x) = -\frac{C}{|x|}, \qquad C = \frac{e^2}{4\pi\varepsilon_0},$$

with the vacuum permittivity ε_0. A classical electron can have any energy $|p|^2/2m - C/|x|$, however negative, by staying close to the proton. A quantum electron cannot. A state squeezed close to the proton has a large mean kinetic energy, and this lecture proves that the mean energy of every state in the domain of the Hamiltonian has a lower bound.

The constants $\hbar$, m and C combine into one length and one energy, the *Bohr radius* and the *Rydberg energy*,

$$a_0 = \frac{\hbar^2}{mC} \approx 0.529\,\text{Å}, \qquad \mathrm{Ry} = \frac{mC^2}{2\hbar^2} = \frac{\hbar^2}{2ma_0^2} \approx 13.61\,\text{eV},$$

so that $C/a_0 = 2\,\mathrm{Ry}$. The potential is unbounded near the origin, so multiplication by V is not a bounded operator, and $H_0 + V$ needs a domain. We show that the vectors of $D(H_0)$ (Definition 13.1) are bounded functions, so that V acts on them.

Lemma 14.1 (The domain of the free energy)**.**

1. *For every* $\psi \in D(H_0)$ *there are* $\psi_k \in \mathcal{S}(\mathbb{R}^3)$ *with* $\psi_k \to \psi$ *and* $H_0\psi_k \to H_0\psi$.
2. *Every* $\psi \in D(H_0)$ *has a continuous representative, and for every* $\eta > 0$

$$\sup_x |\psi(x)| \le \frac{1}{2\sqrt{2\pi}\,\sqrt{\eta}}\left(\frac{2m}{\hbar^2}\,\|H_0\psi\| + \eta^2\,\|\psi\|\right).$$

3. *Let* $\varphi, \chi \in L^2(\mathbb{R}^3)$ *with* $\langle H_0\psi, \varphi\rangle = \langle \psi, \chi\rangle$ *for every* $\psi \in \mathcal{S}(\mathbb{R}^3)$*. Then* $\varphi \in D(H_0)$ *and* $H_0\varphi = \chi$.

Proof.

1. Let $\eta > 0$. Since $\psi \in D(H_0)$, the function $\chi = (|p|^2 + \hbar^2\eta^2)\tilde{\psi}$ is square integrable. By the standing fact S4 there are smooth χ_k that vanish outside bounded sets, with $\chi_k \to \chi$. Put $\tilde{\psi}_k = \chi_k/(|p|^2 + \hbar^2\eta^2)$ and $\psi_k = F^{-1}\tilde{\psi}_k$. By the standing fact S1,

$$\psi_k(x) = (2\pi\hbar)^{-3/2} \int e^{\mathrm{i}p\cdot x/\hbar} \tilde{\psi}_k(p)\, \mathrm{d}^3p.$$

   Differentiating under the integral multiplies $\tilde{\psi}_k$ by $\mathrm{i}p_j/\hbar$, and multiplying by x_j differentiates it, after an integration by parts in p_j as in Proposition 4.3. So every product of a monomial with a partial derivative of ψ_k is such an integral of a smooth function that vanishes outside a ball, and it is bounded. Hence $\psi_k \in \mathcal{S}(\mathbb{R}^3)$. Since $|p|^2/(|p|^2 + \hbar^2\eta^2) \le 1$, the unitarity of F gives $\|H_0(\psi_k - \psi)\| \le \|\chi_k - \chi\|/2m$ and $\|\psi_k - \psi\| \le \|\chi_k - \chi\|/\hbar^2\eta^2$, and both tend to 0.

2. Take ψ_k and χ_k as in the first step, for this η. Let N be the norm of $1/(|p|^2 + \hbar^2\eta^2)$, so that

$$N^2 = \int_0^\infty \frac{4\pi p^2\, \mathrm{d}p}{(p^2 + \hbar^2\eta^2)^2} = \frac{\pi^2}{\hbar\eta}.$$

   By the Cauchy–Schwarz inequality,

$$|\psi_k(x) - \psi_{k'}(x)| \le (2\pi\hbar)^{-3/2} \int \left|\tilde{\psi}_k - \tilde{\psi}_{k'}\right| \mathrm{d}^3p \le (2\pi\hbar)^{-3/2} N \|\chi_k - \chi_{k'}\| \quad \text{for every } x.$$

   So the ψ_k converge uniformly to a continuous function f. On every cube K, $\int_K |\psi_k - f|^2$ tends to 0, and so does $\int_K |\psi_k - \psi|^2$. So ψ and f agree on every cube, and f represents ψ. Two continuous functions that agree on every cube are equal, so f does not depend on η. In the same way $|\psi_k(x)| \le (2\pi\hbar)^{-3/2} N \|\chi_k\|$, and in the limit $|f(x)| \le (2\pi\hbar)^{-3/2} N \|\chi\|$, with $\|\chi\| \le 2m \|H_0\psi\| + \hbar^2\eta^2 \|\psi\|$. Since $(2\pi\hbar)^{-3/2}\pi/\sqrt{\hbar\eta} = 1/(2\sqrt{2\pi}\,\hbar^2\sqrt{\eta})$, this is the bound.

3. Let $\psi \in D(H_0)$, with ψ_k as in the first step. Then $\langle H_0\psi, \varphi\rangle = \lim \langle H_0\psi_k, \varphi\rangle = \lim \langle \psi_k, \chi\rangle = \langle \psi, \chi\rangle$. Taking complex conjugates, $\langle \varphi, H_0\psi\rangle = \langle \chi, \psi\rangle$ for every $\psi \in D(H_0)$. By Definition 0.2, $\varphi \in D(H_0^\dagger)$ with $H_0^\dagger\varphi = \chi$, and $H_0^\dagger = H_0$ by Proposition 13.2. ∎

The bound holds because $1/(|p|^2 + \hbar^2\eta^2)$ is square integrable in three dimensions. With it, we bound $V\psi$ by $H_0\psi$ and ψ.

Theorem 14.2 (The Hamiltonian of hydrogen). *Every $\psi \in D(H_0)$ has $V\psi \in L^2(\mathbb{R}^3)$, with*

$$\|V\psi\| \le \tfrac{1}{2} \|H_0\psi\| + 11\,\mathrm{Ry}\, \|\psi\|,$$

and $H = H_0 + V$, on $D(H) = D(H_0)$, is self-adjoint. For every $\psi \in D(H)$ there are $\psi_k \in \mathcal{S}(\mathbb{R}^3)$ with $\psi_k \to \psi$ and $H\psi_k \to H\psi$.

Proof. First, the bound. Let $\rho > 0$, and take the continuous representative of ψ. Since $\int_{|x|<\rho} |x|^{-2}\, \mathrm{d}^3x = 4\pi\rho$, and $|x|^{-2} \le \rho^{-2}$ for $|x| \ge \rho$,

$$\|V\psi\|^2 = C^2 \int \frac{|\psi(x)|^2}{|x|^2}\, \mathrm{d}^3x \le C^2 \Big(4\pi\rho \sup|\psi|^2 + \frac{\|\psi\|^2}{\rho^2}\Big).$$

Since $\sqrt{a+b} \le \sqrt{a} + \sqrt{b}$ for $a, b \ge 0$, it follows that $\|V\psi\| \le C\sqrt{4\pi\rho} \sup|\psi| + C\|\psi\|/\rho$. With $\rho = 1/\eta$, the second claim of Lemma 14.1 and $mC/\hbar^2 = 1/a_0$ give

$$\|V\psi\| \le \frac{\sqrt{2}}{a_0\eta} \|H_0\psi\| + \Big(1 + \frac{1}{\sqrt{2}}\Big) C\eta \|\psi\|.$$

For $\eta = 3/a_0$ the first coefficient is $\sqrt{2}/3 < \frac{1}{2}$, and the second is $6(1 + 1/\sqrt{2})\,\mathrm{Ry} < 11\,\mathrm{Ry}$.

Next, H is symmetric, since H_0 is and V is real. Let $b = 44\,\mathrm{Ry}$, and let $z = \mathrm{i}b$ or $z = -\mathrm{i}b$. For $\varphi \in L^2(\mathbb{R}^3)$, the vector $\psi = (H_0 - z)^{-1}\varphi$ (Theorem 6.3) satisfies $\|\varphi\|^2 = \|H_0\psi\|^2 + b^2\,\|\psi\|^2$, by (6.1). So $\|H_0\psi\| \le \|\varphi\|$ and $\|\psi\| \le \|\varphi\|/b$. Let $K = V(H_0 - z)^{-1}$. By the bound,

$$\|K\varphi\| = \|V\psi\| \le \frac{1}{2}\,\|H_0\psi\| + 11\,\mathrm{Ry}\;\|\psi\| \le \left(\frac{1}{2} + \frac{11\,\mathrm{Ry}}{b}\right)\|\varphi\| = \frac{3}{4}\,\|\varphi\|\,.$$

Let $\chi \in L^2(\mathbb{R}^3)$. The vectors $(-K)^i\chi$ have norms at most $(3/4)^i\,\|\chi\|$, so the partial sums of $\sum_{i\ge 0}(-K)^i\chi$ form a Cauchy sequence, with a limit φ. Since $(\mathbb{1} + K)$ maps the partial sum up to $i = j$ to $\chi - (-K)^{j+1}\chi$, the limit satisfies $\varphi + K\varphi = \chi$. Then $\psi = (H_0 - z)^{-1}\varphi$ lies in $D(H_0)$, and

$$(H - z)\psi = (H_0 - z)\psi + V\psi = \varphi + K\varphi = \chi.$$

So $H - \mathrm{i}b$ and $H + \mathrm{i}b$ map $D(H)$ onto $L^2(\mathbb{R}^3)$, and H is self-adjoint by Proposition 6.7.

Finally, for $\psi \in D(H)$ take ψ_k as in the first claim of Lemma 14.1. By the bound, $\|V(\psi_k - \psi)\| \le \frac{1}{2}\,\|H_0(\psi_k - \psi)\| + 11\,\mathrm{Ry}\;\|\psi_k - \psi\| \to 0$. ∎

This is the Kato–Rellich theorem for the Coulomb potential. The proof needs the coefficient $\frac{1}{2}$ of $\|H_0\psi\|$ in the bound to be below 1. For large b the bound then gives $\|K\varphi\| \le q\,\|\varphi\|$ with some $q < 1$, so the series $\sum_i(-K)^i\chi$ converges. A coefficient 1 or more would give $q \ge 1$ for every b.

Self-check 14.1

1. Let $\rho > 0$ and $\psi = (3/4\pi\rho^3)^{1/2}$ on $|x| < \rho$, and $\psi = 0$ outside. Show that $\|\psi\| = 1$ and $\|V\psi\| = \sqrt{3}\,C/\rho$. Why is ψ not in $D(H_0)$?
2. The proof of Theorem 14.2 takes $b = 44\,\mathrm{Ry}$ and shows that $\|K\varphi\| \le q\,\|\varphi\|$ with $q = \frac{1}{2} + 11\,\mathrm{Ry}/b$. For which $b > 0$ is this q below 1? For such b, which earlier criterion proves that H is self-adjoint?

14.2 Hydrogen is stable

We now prove the lower bound on the mean energy. To show that it is attained we need the action of H_0 on functions that are not smooth at the origin, such as $e^{-|x|/a_0}$.

Lemma 14.3 (Functions smooth away from the origin). *Let ψ be a bounded continuous function on $\mathbb{R}^3$ that is square integrable, twice continuously differentiable on $x \ne 0$, with bounded first partial derivatives there, and suppose that $\Delta\psi$, computed on $x \ne 0$, is square integrable. Then $\psi \in D(H_0)$ and $H_0\psi = -(\hbar^2/2m)\Delta\psi$.*

Proof. Let $\varphi \in \mathcal{S}(\mathbb{R}^3)$ and $0 < \epsilon < R$. The boundary of the shell $\epsilon < |x| < R$ consists of two spheres (Figure 14.1, left). Its outward normal is $\hat{x} = x/|x|$ on the outer sphere $|x| = R$, and $-\hat{x}$ on the inner sphere $|x| = \epsilon$, where it points towards the origin. So the divergence theorem, applied on the shell to $\overline{\varphi}\nabla\psi - \psi\nabla\overline{\varphi}$, whose divergence is $\overline{\varphi}\Delta\psi - \psi\Delta\overline{\varphi}$, adds the flux along $\hat{x}$ through the outer sphere and subtracts the flux along $\hat{x}$ through the inner one:

$$\int_{\epsilon<|x|<R} \left(\overline{\varphi}\Delta\psi - \psi\Delta\overline{\varphi}\right) \mathrm{d}^3x = \left[\int_{|x|=s} \left(\overline{\varphi}\,\partial_r\psi - \psi\,\partial_r\overline{\varphi}\right) \mathrm{d}S\right]_{s=\epsilon}^{s=R},$$

where ∂_r is the derivative along $\hat{x}$. Let M bound $|\psi|$ and $|\nabla\psi|$. The term at $s = \epsilon$ is at most $4\pi\epsilon^2 M \sup(|\varphi| + |\nabla\varphi|)$, which tends to 0. The term at $s = R$ is at most $4\pi R^2 M$ times the maximum of $|\varphi| + |\nabla\varphi|$ on $|x| = R$, which tends to 0 since $\varphi \in \mathcal{S}(\mathbb{R}^3)$. The products $\overline{\varphi}\Delta\psi$ and

$\psi\Delta\overline{\varphi}$ are integrable, as products of two square integrable functions. So $\langle\varphi, \Delta\psi\rangle = \langle\Delta\varphi, \psi\rangle$. Since $H_0\varphi = -(\hbar^2/2m)\Delta\varphi$ (Proposition 13.2), this says that $\langle H_0\varphi, \psi\rangle = \langle\varphi, -(\hbar^2/2m)\Delta\psi\rangle$ for every $\varphi \in \mathcal{S}(\mathbb{R}^3)$. The third claim of Lemma 14.1 gives the result. ∎

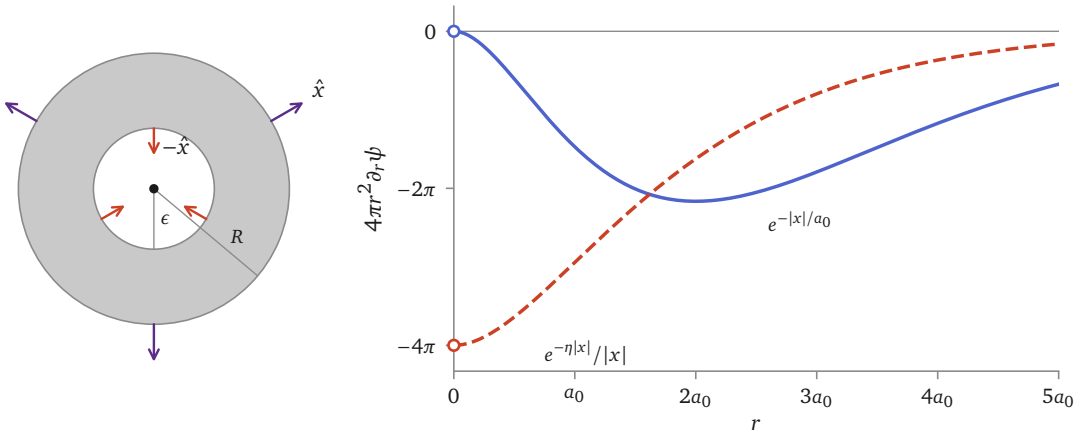


Figure 14.1: Left: a cross-section of the shell $\epsilon < |x| < R$ and the outward normals of its boundary, $\hat{x}$ on the outer sphere and $-\hat{x}$ on the inner sphere. Right: the flux $4\pi r^2\partial_r\psi$ of $\nabla\psi$ along $\hat{x}$ through the sphere $|x| = r$, for $\psi = e^{-|x|/a_0}$ (solid) and for $\psi = e^{-\eta|x|}/|x|$ with $\eta = 1/a_0$ (dashed), with lengths in units of a_0. As $r \to 0$, the first flux tends to 0 and the second to -4π.

The conditions at the origin cannot be dropped. Let $\eta > 0$ and $\psi(x) = e^{-\eta|x|}/|x|$. Then $\Delta\psi = \eta^2\psi$ on $x \neq 0$, and ψ is square integrable, since $\int 4\pi r^2 |\psi|^2 \, \mathrm{d}r = 2\pi/\eta$. But ψ is unbounded near 0, so by Lemma 14.1 it does not lie in $D(H_0)$. The rule on $x \neq 0$ misses what happens at the origin. The flux of $\nabla\psi$ through a small sphere about 0 tends to -4π, as for a point source at the origin, like the δ of Lecture 11 in three dimensions. For $e^{-|x|/a_0}$ the same flux tends to 0 (Figure 14.1, right). So in the proof of Lemma 14.3 the term at $s = \epsilon$ tends to $-4\pi\overline{\varphi(0)}$ for ψ, not to 0.

Theorem 14.4 (Hydrogen is stable). *For every $\psi \in D(H)$,*

$$\langle\psi, H\psi\rangle \geq -\mathrm{Ry}\, \|\psi\|^2 .$$

So $\sigma(H) \subseteq [-\mathrm{Ry}, \infty)$. Moreover, $-\mathrm{Ry}$ is an eigenvalue of H, with the eigenvector $\psi_1(x) = e^{-|x|/a_0}$.

Proof. We write $\langle\psi, (H + \mathrm{Ry})\psi\rangle$ on $\mathcal{S}(\mathbb{R}^3)$ as a sum of squared norms. The approximation of Theorem 14.2 then extends the inequality to $D(H)$.

First let $\psi \in \mathcal{S}(\mathbb{R}^3)$, and put

$$A_j\psi = P_j\psi - \mathrm{i}\,\frac{\hbar}{a_0}\,\frac{x_j}{|x|}\,\psi, \qquad j = 1, 2, 3,$$

a square-integrable function, since $|x_j| \leq |x|$. Expanding the squares, with $\sum_j x_j^2/|x|^2 = 1$ and $\sum_j \|P_j\psi\|^2 = 2m\,\langle\psi, H_0\psi\rangle$,

$$0 \leq \sum_j \|A_j\psi\|^2 = 2m\,\langle\psi, H_0\psi\rangle + \frac{\hbar^2}{a_0^2}\,\|\psi\|^2 + \frac{\hbar^2}{a_0}\int \frac{x}{|x|}\cdot\nabla|\psi|^2 \, \mathrm{d}^3x,$$

since the cross terms are

$$2\,\mathrm{Re}\left\langle -\mathrm{i}\hbar\partial_j\psi, -\mathrm{i}\frac{\hbar}{a_0}\frac{x_j}{|x|}\psi\right\rangle = \frac{\hbar^2}{a_0}\int \frac{x_j}{|x|}\,\partial_j\,|\psi|^2\;\mathrm{d}^3x.$$

On $x \neq 0$ the divergence of $x/|x|$ is $2/|x|$. By the divergence theorem on the shell $\epsilon < |x| < R$, as in the proof of Lemma 14.3,

$$\int \frac{x}{|x|}\cdot\nabla\,|\psi|^2\;\mathrm{d}^3x = -2\int \frac{|\psi|^2}{|x|}\,\mathrm{d}^3x,$$

because the term on the small sphere is at most $4\pi\epsilon^2 \sup|\psi|^2$ and vanishes as $\epsilon \to 0$, and the term on the large sphere vanishes as $R \to \infty$. Dividing by $2m$, with $\hbar^2/ma_0 = C$ and $\hbar^2/2ma_0^2 = \mathrm{Ry}$,

$$\langle\psi, H_0\psi\rangle - C\left\langle\psi, |x|^{-1}\,\psi\right\rangle + \mathrm{Ry}\;\|\psi\|^2 \geq 0,$$

which is the claim for $\psi \in \mathcal{S}(\mathbb{R}^3)$. For $\psi \in D(H)$, take ψ_k as in Theorem 14.2. Then $\langle\psi_k, H\psi_k\rangle \to \langle\psi, H\psi\rangle$ and $\|\psi_k\| \to \|\psi\|$, and the claim follows in the limit.

Next, by Proposition 10.3, the least point of $\sigma(H)$ is the infimum of the means $\langle\psi, H\psi\rangle$ over unit vectors, which is at least $-\mathrm{Ry}$.

Finally, the bound is attained at ψ_1. The rule A_j is chosen for this. On $x \neq 0$, $-\mathrm{i}\hbar\partial_j\psi_1 = \mathrm{i}(\hbar/a_0)(x_j/|x|)\psi_1$, so $A_j\psi_1 = 0$ there, as $ah_0 = 0$ for the oscillator of Lecture 8. Since ψ_1 does not lie in $\mathcal{S}(\mathbb{R}^3)$, we check the eigenvalue directly. The function ψ_1 is bounded, continuous and square integrable. On $x \neq 0$, $\nabla\psi_1 = -(x/a_0\,|x|)\psi_1$ is bounded, and

$$\Delta\psi_1 = \Big(\frac{1}{a_0^2} - \frac{2}{a_0\,|x|}\Big)\psi_1,$$

which is square integrable, since $\int_{|x|<a_0} |x|^{-2}\;\mathrm{d}^3x = 4\pi a_0$. By Lemma 14.3, $\psi_1 \in D(H)$ and $H_0\psi_1 = -(\hbar^2/2m)\Delta\psi_1 = -\mathrm{Ry}\,\psi_1 + (C/|x|)\psi_1$. So $H\psi_1 = -\mathrm{Ry}\,\psi_1$. ∎

Figure 14.2 shows how the bound arises on the states of $e^{-|x|/a}$, with $a > 0$, which are squeezed towards the proton as a shrinks. As for ψ_1 in the proof, Lemma 14.3 applies, with $\Delta e^{-|x|/a} = (1/a^2 - 2/a\,|x|)e^{-|x|/a}$ on $x \neq 0$. Integrating in spherical coordinates, the mean of $|x|^{-1}$ in this state is $1/a$. So the mean of H_0 is $-(\hbar^2/2m)(1/a^2 - 2/a^2) = \hbar^2/2ma^2 = \mathrm{Ry}\,a_0^2/a^2$, and the mean of V is $-C/a = -2\,\mathrm{Ry}\,a_0/a$. As a shrinks, the mean of V falls like $1/a$, but the mean of H_0 grows like $1/a^2$. Their sum, $\mathrm{Ry}(a_0^2/a^2 - 2a_0/a)$, is least at $a = a_0$, where it equals the bound $-\mathrm{Ry}$.

The bound needs the domain. The function $e^{-\eta|x|}/|x|$ above satisfies the free rule with the energy $-\hbar^2\eta^2/2m$, as low as one likes, away from the origin. On the smooth functions that vanish near the origin, the rule of H has other self-adjoint extensions. They differ from H only at angular momentum 0, by a point source at the origin, and each has a level below $-\mathrm{Ry}$ (Albeverio et al., Thms I.2.1.1 and I.2.1.3). The domain $D(H_0)$, whose functions are bounded, excludes them.

Self-check 14.2

1. Let $a > 0$ and $\psi = e^{-|x|/a}$. Show that $\|\psi\|^2 = \pi a^3$ and $\left\langle\psi, |x|^{-1}\,\psi\right\rangle = \pi a^2$, so that the mean of $|x|^{-1}$ in the state of ψ is $1/a$.

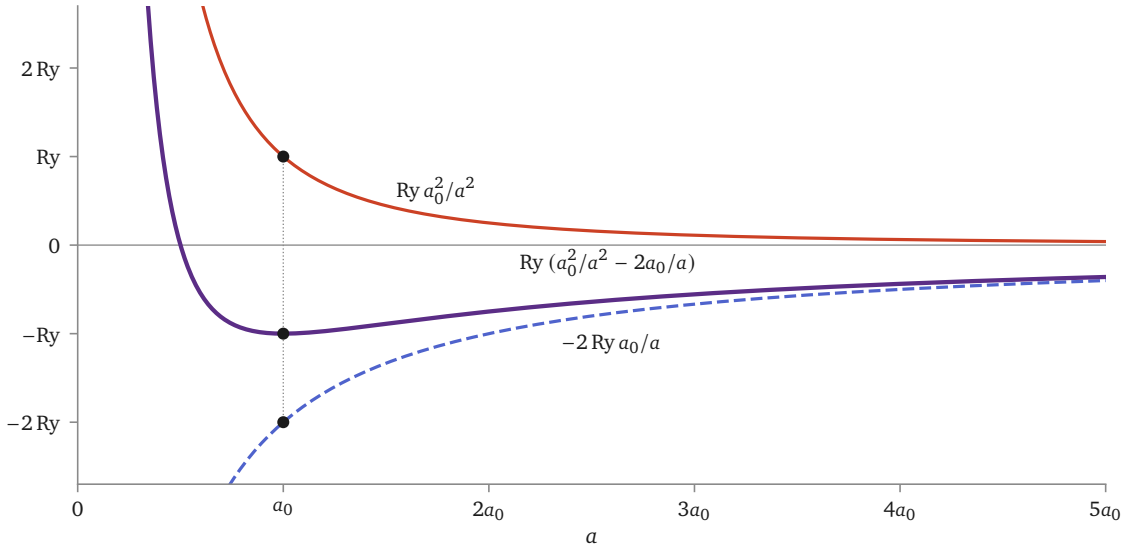


Figure 14.2: The means in the state of $e^{-|x|/a}$, against a: of H_0 (upper solid curve), of V (dashed) and of H (lower solid curve). The dashed curve falls without bound as $a \to 0$, the upper curve rises faster, and their sum is least at $a = a_0$, where it is $-\mathrm{Ry}$. The marks at $a = a_0$ are the means Ry, $-2\,\mathrm{Ry}$ and $-\mathrm{Ry}$ of the ground state $e^{-|x|/a_0}$.

14.3 The continuum and the discrete spectrum

An electron far from the proton is nearly free, and the spectrum of H contains that of H_0. We show this with packets that widen and move away from the proton. Let $p_0 \geq 0$, and let f be a real smooth function on $\mathbb{R}^3$ that vanishes outside the unit ball, with $\|f\| = 1$. For $k \geq 2$ put $d_k = (k^2, 0, 0)$ and

$$\psi_k(x) = k^{-3/2} f\Big(\frac{x - d_k}{k}\Big) e^{\mathrm{i}p_0 x_1/\hbar}.$$

The packet ψ_k vanishes outside the ball of radius k about d_k (Figure 14.3). The radius grows, so the derivatives of the profile f contribute less and less. The centre moves away faster than the radius grows, so the potential on the packet tends to 0.

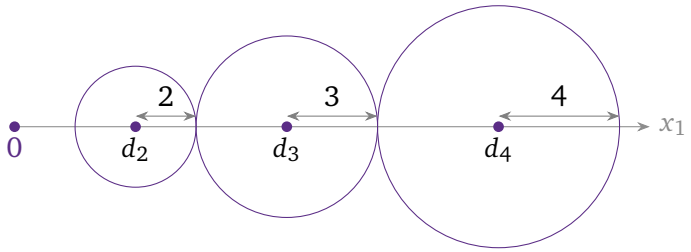


Figure 14.3: Cross-sections of the balls of radius k about $d_k = (k^2, 0, 0)$, for $k = 2, 3, 4$. The packet ψ_k vanishes outside its ball. Neighbouring balls touch at one point, and their interiors are disjoint. The distance from the origin, where V is singular, to the ball of radius k is $k^2 - k$.

Proposition 14.5 (The continuum). *Every $\lambda \geq 0$ lies in $\sigma(H)$, and $[0,\infty) \subseteq \sigma_{\mathrm{ess}}(H)$.*

Proof. Let $\lambda = p_0^2/2m$ with $p_0 \geq 0$, and take the packets ψ_k above for this p_0. Each is a unit vector in $\mathcal{S}(\mathbb{R}^3)$, as in the proof of Theorem 10.10. By the product rule,

$$(H_0 - \lambda)\psi_k = -\frac{\hbar^2}{2m}\, k^{-3/2}\Big(\frac{\Delta f}{k^2} + \frac{2\mathrm{i}p_0}{\hbar k}\,\partial_1 f\Big)\Big(\frac{x-d_k}{k}\Big)e^{\mathrm{i}p_0 x_1/\hbar},$$

and the substitution $x = d_k + ky$ gives $\|(H_0-\lambda)\psi_k\| \leq (\hbar^2/2m)\big(\|\Delta f\|/k^2 + 2p_0\,\|\partial_1 f\|/\hbar k\big) \to 0$. Where $\psi_k \neq 0$, $|x| \geq |d_k| - |x-d_k| \geq k^2 - k \geq k^2/2$, so $\|V\psi_k\| \leq 2C/k^2$. Hence $\|(H-\lambda)\psi_k\| \to 0$, and $\lambda \in \sigma(H)$ by Proposition 6.2. No point of the interval $[0,\infty)$ is isolated in $\sigma(H)$, so $[0,\infty) \subseteq \sigma_{\mathrm{ess}}(H)$ (Definition 6.9). ∎

Physically, an electron with an energy $E \geq 0$ is not bound to the proton, and the atom is *ionised*. By Theorem 14.4 and Proposition 14.5, the spectrum lies in $[-\mathrm{Ry},\infty)$ and contains $[0,\infty)$. So it takes at least $\mathrm{Ry} \approx 13.6\,\mathrm{eV}$ to ionise hydrogen from the state of ψ_1.

We show next that below 0 the spectrum is discrete. As in Lecture 10, the proof uses local compactness, now in three dimensions.

Lemma 14.6 (Local compactness in space). *Let $\psi_k \in D(H_0)$ with $\|\psi_k\| \leq M$ and $\|H_0\psi_k\| \leq M$ for every k. Then a subsequence of (ψ_k) converges uniformly on every bounded set.*

Proof. Fix $\eta > 0$. By the second claim of Lemma 14.1, the ψ_k are continuous and bounded by one constant. Let $\psi \in D(H_0)$, let x and y be points with $d = |x-y| > 0$, and take the approximants of ψ built in the proof of Lemma 14.1 for this η, written φ_j, with $\chi_j \to \chi = (|p|^2 + \hbar^2\eta^2)\tilde{\psi}$. They converge to ψ uniformly (the second step of that proof), and as there,

$$\big|\varphi_j(x) - \varphi_j(y)\big| \leq (2\pi\hbar)^{-3/2} N_d\, \big\|\chi_j\big\|, \qquad N_d^2 = \int \frac{\big|e^{\mathrm{i}p\cdot(x-y)/\hbar} - 1\big|^2}{(|p|^2+\hbar^2\eta^2)^2}\,\mathrm{d}^3p.$$

Since $\big|e^{\mathrm{i}\theta} - 1\big| \leq \min(|\theta|, 2)$, $|p\cdot(x-y)| \leq |p|\,d$ and $(|p|^2+\hbar^2\eta^2)^2 \geq |p|^4$, splitting the integral at $|p| = \hbar/d$ gives

$$N_d^2 \leq \frac{d^2}{\hbar^2}\int_{|p|<\hbar/d} \frac{\mathrm{d}^3p}{|p|^2} + 4\int_{|p|>\hbar/d} \frac{\mathrm{d}^3p}{|p|^4} = \frac{20\pi d}{\hbar}.$$

In the limit,

$$|\psi(x) - \psi(y)| \leq (2\pi\hbar)^{-3/2}\Big(\frac{20\pi\,|x-y|}{\hbar}\Big)^{1/2}\big(2m\,\|H_0\psi\| + \hbar^2\eta^2\,\|\psi\|\big).$$

So the ψ_k are bounded and equicontinuous with one constant. The diagonal argument of the proof of Lemma 10.9, with the points of $\mathbb{R}^3$ that have rational coordinates, gives a subsequence that converges at each of them. On every ball, finitely many of these points lie within any given distance of every point of the ball, and as there the equicontinuity makes the subsequence converge uniformly on the ball. ∎

Theorem 14.7 (The spectrum below 0). *Every point of $\sigma(H)$ below 0 is an eigenvalue in $\sigma_{\mathrm{d}}(H)$. So $\sigma_{\mathrm{ess}}(H) = [0,\infty)$.*

Proof. First, let $\lambda < 0$, and let $\psi_k \in D(H)$ be unit vectors with $(H-\lambda)\psi_k \to 0$. We show that a subsequence converges to a unit vector ψ with $H\psi = \lambda\psi$. Since $H_0\psi_k = H\psi_k - V\psi_k$, Theorem 14.2 gives $\|H_0\psi_k\| \le \|H\psi_k\| + \frac{1}{2}\|H_0\psi_k\| + 11\,\mathrm{Ry}$, so $\|H_0\psi_k\| \le 2\|H\psi_k\| + 22\,\mathrm{Ry}$, which is bounded. By Lemma 14.6, a subsequence, still written (ψ_k), converges uniformly on every ball, and by Lemma 14.1 the ψ_k are bounded by one constant B. For $0 < \rho < R$, splitting the integral at $|x| = \rho$ and $|x| = R$ as in the proof of Theorem 14.2,

$$\|V(\psi_k - \psi_{k'})\| \le 2BC\sqrt{4\pi\rho} + \frac{2C}{R} + \frac{C}{\rho}\Big(\frac{4\pi R^3}{3}\Big)^{1/2} \sup_{|x|\le R} |\psi_k(x) - \psi_{k'}(x)|\,.$$

Choosing ρ small, R large, and then k and k' large makes this small, so $(V\psi_k)$ converges. Since $\lambda < 0$, the operator $H_0 - \lambda$ has the bounded inverse $F^{-1}M_{1/(|p|^2/2m-\lambda)}F$, as in Proposition 6.8. So $\psi_k = (H_0-\lambda)^{-1}\big((H-\lambda)\psi_k - V\psi_k\big)$ converges, to a unit vector ψ. For every $\varphi \in D(H)$, $\langle (H-\lambda)\varphi, \psi\rangle = \lim \langle \varphi, (H-\lambda)\psi_k\rangle = 0$. By Definition 0.2 and Theorem 14.2, $\psi \in D(H)$ and $H\psi = \lambda\psi$.

Next, let $\lambda < 0$ lie in $\sigma(H)$. By Weyl's criterion (Theorem 6.3) the first step applies, and λ is an eigenvalue. Its eigenspace has finite dimension, since an infinite orthonormal sequence of eigenvectors would satisfy the hypothesis of the first step, while $\|\psi_k - \psi_{k'}\|^2 = 2$ for $k \ne k'$. As in the proof of Theorem 10.10, λ is isolated: otherwise there are points $\lambda_k \ne \lambda$ of $\sigma(H)$ with $\lambda_k \to \lambda$, negative and so eigenvalues, and unit eigenvectors for distinct ones are orthonormal, satisfy the hypothesis of the first step, and have no convergent subsequence. So $\lambda \in \sigma_{\mathrm{d}}(H)$. With Proposition 14.5, $\sigma_{\mathrm{ess}}(H) = [0,\infty)$. ∎

In general this is Weyl's theorem, as in the remark after Theorem 10.10. Next, no eigenvalue lies at or above 0. The proof uses the *dilations* $(T_\gamma\psi)(x) = e^{3\gamma/2}\psi(e^\gamma x)$, for real γ, which are unitary, since the change of variables $y = e^\gamma x$ has the Jacobian $e^{3\gamma}$.

Proposition 14.8 (The virial theorem). *If $H\psi = E\psi$ with $\psi \ne 0$, then $2\langle\psi, H_0\psi\rangle + \langle\psi, V\psi\rangle = 0$, and $E\|\psi\|^2 = -\langle\psi, H_0\psi\rangle < 0$.*

Proof. First, the scaling. For $\psi \in \mathcal{S}(\mathbb{R}^3)$ the chain rule gives $\Delta(T_\gamma\psi) = e^{2\gamma}T_\gamma(\Delta\psi)$, so $H_0T_\gamma\psi = e^{2\gamma}T_\gamma H_0\psi$. For $\varphi \in \mathcal{S}(\mathbb{R}^3)$ and $\psi \in D(H_0)$, since $T_\gamma^\dagger = T_{-\gamma}$ and H_0 is symmetric,

$$\big\langle H_0\varphi, T_\gamma\psi\big\rangle = e^{2\gamma}\big\langle H_0T_{-\gamma}\varphi, \psi\big\rangle = e^{2\gamma}\big\langle T_{-\gamma}\varphi, H_0\psi\big\rangle = \big\langle \varphi, e^{2\gamma}T_\gamma H_0\psi\big\rangle\,.$$

By the third claim of Lemma 14.1, $T_\gamma\psi \in D(H_0)$ and $H_0T_\gamma\psi = e^{2\gamma}T_\gamma H_0\psi$. Since $V(x) = e^\gamma V(e^\gamma x)$, also $VT_\gamma\psi = e^\gamma T_\gamma V\psi$.

Next, let $H\psi = E\psi$, and put $f_1(\gamma) = \big\langle\psi, T_\gamma H_0\psi\big\rangle$, $f_2(\gamma) = \big\langle\psi, T_\gamma V\psi\big\rangle$ and $f_0(\gamma) = \big\langle\psi, T_\gamma\psi\big\rangle$. Since H is symmetric and E is real,

$$Ef_0(\gamma) = \big\langle H\psi, T_\gamma\psi\big\rangle = \big\langle\psi, HT_\gamma\psi\big\rangle = e^{2\gamma}f_1(\gamma) + e^\gamma f_2(\gamma), \qquad Ef_0(\gamma) = \big\langle\psi, T_\gamma H\psi\big\rangle = f_1(\gamma) + f_2(\gamma).$$

Subtracting and dividing by $e^\gamma - 1$, for $\gamma \ne 0$, $(e^\gamma+1)f_1(\gamma) + f_2(\gamma) = 0$. For a continuous φ that vanishes outside a ball, $T_\gamma\varphi \to \varphi$ uniformly on a ball as $\gamma \to 0$, so in norm. By the standing fact S4 and $\|T_\gamma\| = 1$, $T_\gamma\varphi \to \varphi$ for every φ. So f_1 and f_2 are continuous at 0, and letting $\gamma \to 0$, this gives $2f_1(0) + f_2(0) = 0$.

Finally,

$$E\|\psi\|^2 = \langle\psi, H_0\psi\rangle + \langle\psi, V\psi\rangle = -\langle\psi, H_0\psi\rangle = -\int \frac{|p|^2}{2m}\big|\tilde\psi(p)\big|^2\,\mathrm{d}^3p.$$

If this vanished, $\tilde\psi$ would vanish on every cube that misses the origin, and $\psi = 0$, as in the proof of Proposition 13.2. So $E < 0$. ∎

For the ground state the identity can be seen directly. By the means computed before Figure 14.2, at $a = a_0$ the state of ψ_1 has the mean Ry of H_0 and the mean -2 Ry of V, the marks of the figure. So $2\,\mathrm{Ry} - 2\,\mathrm{Ry} = 0$, and the eigenvalue $-\mathrm{Ry}$ is minus the mean of H_0.

Self-check 14.3

1. Show that the packets ψ_k defined before Proposition 14.5 are orthogonal for $2 \le k < k'$.

The physics

With a fixed proton, the energy needed to ionise hydrogen from the state of ψ_1 is the distance from $-\mathrm{Ry}$ to 0. For the mass m of the electron this is $\mathrm{Ry} \approx 13.6057$ eV. NIST's value of the ionisation energy is 13.5984 eV. The proton moves too. With the reduced mass $mm_p/(m+m_p)$ in place of m, the Rydberg energy becomes 13.5983 eV, only about 0.00015 eV below NIST's value. This difference comes from relativistic and radiative effects, which this model leaves out.

Hydrogen: the Hamiltonian

States: the electron's states are the rays represented by unit vectors of $L^2(\mathbb{R}^3)$; every vector of $D(H_0)$ is a bounded continuous function (Lemma 14.1). *Observables:* $H = H_0 - C/|x|$ is self-adjoint on $D(H_0)$ (Theorem 14.2), and $\langle \psi, H\psi\rangle \ge -\mathrm{Ry}\,\|\psi\|^2$ for every $\psi \in D(H)$ (Theorem 14.4). *Measurement:* a reading of the energy in an interval below $-\mathrm{Ry}$ answers yes with probability 0 (Theorem 14.4, and Theorem 6.4, cited).

Summary.

- *The Hamiltonian of hydrogen*: every vector of $D(H_0)$ is a bounded continuous function (Lemma 14.1), and $H_0 - C/|x|$ is self-adjoint on $D(H_0)$ (Theorem 14.2).
- *Hydrogen is stable*: $\langle \psi, H\psi\rangle \ge -\mathrm{Ry}\,\|\psi\|^2$ on $D(H)$, with equality for $e^{-|x|/a_0}$ (Theorem 14.4). The domain matters: $e^{-\eta|x|}/|x|$ satisfies the free rule with the energy $-\hbar^2\eta^2/2m$ away from the origin, and lies outside $D(H_0)$ (§ 14.2).
- *The continuum*: $\sigma_{\mathrm{ess}}(H) = [0, \infty)$, the spectrum below 0 is discrete, and every eigenvalue is negative (Propositions 14.5 and 14.8 and Theorem 14.7).

Lecture 15 finds the eigenvalues and their eigenvectors.

Self-check answers.

14.1 1. $\|\psi\|^2 = (3/4\pi\rho^3)(4\pi\rho^3/3) = 1$ and $\|V\psi\|^2 = C^2(3/4\pi\rho^3)\int_0^\rho 4\pi\,\mathrm{d}r = 3C^2/\rho^2$. Every vector of $D(H_0)$ is continuous (Lemma 14.1), and ψ has no continuous representative. 2. As in the proof, $\|H_0\psi\| \le \|\varphi\|$ and $\|\psi\| \le \|\varphi\|/b$ for $\psi = (H_0 \mp ib)^{-1}\varphi$, so the bound gives $\|K\varphi\| \le q\,\|\varphi\|$ with $q = \frac{1}{2} + 11\,\mathrm{Ry}/b$. This q is below 1 exactly when $b > 22\,\mathrm{Ry}$, and $b = 44\,\mathrm{Ry}$ gives $q = \frac{3}{4}$. With $q < 1$ the series $\sum_i (-K)^i\chi$ converges, so $H - ib$ and $H + ib$ map $D(H)$ onto $L^2(\mathbb{R}^3)$, and Proposition 6.7 makes the symmetric operator H self-adjoint.

14.2 1. In spherical coordinates, $\|\psi\|^2 = \int_0^\infty 4\pi r^2 e^{-2r/a}\,\mathrm{d}r = 4\pi\cdot 2(a/2)^3 = \pi a^3$ and $\left\langle \psi, |x|^{-1}\,\psi\right\rangle = \int_0^\infty 4\pi r e^{-2r/a}\,\mathrm{d}r = 4\pi(a/2)^2 = \pi a^2$. The mean of $|x|^{-1}$ is the quotient $\pi a^2/\pi a^3 = 1/a$.

14.3 1. The function ψ_k vanishes outside the ball of radius k about d_k. The centres are $k'^2 - k^2 = (k'-k)(k'+k) \ge k + k'$ apart, so the two balls meet at most in a point, and $\overline{\psi_k}\psi_{k'} = 0$ elsewhere. So $\langle \psi_k, \psi_{k'}\rangle = 0$.

References

Teschl, *Mathematical Methods in Quantum Mechanics*, Lemma 10.1 and Thm 10.2 (pp. 221–222), Thm 10.3 (p. 223), Thm 6.4 (p. 135). The bound of Lemma 14.1 is Teschl's Lemma 10.1. His Thm 6.4 is the Kato–Rellich theorem, and his Thm 10.2 applies it to the Coulomb potential, as Theorem 14.2 does. His Thm 10.2 also obtains the essential spectrum of Theorem 14.7, by Weyl's theorem. Proposition 14.8 is his Thm 10.3, with his proof. Hall, *Quantum Theory for Mathematicians*, Prop. 9.35 (p. 189). S. Albeverio et al., *Solvable Models in Quantum Mechanics*, Thms I.2.1.1 and I.2.1.3 (pp. 54–56). NIST Atomic Spectra Database, ground states and ionization energies, H I.